Sampled insertion: static qualification

7 October 2026 — E1 approved; Stage 3 remains open

Scope. Conservative reset and local compatibility deductions, followed by a conditional prescribed-path analytical reference. No joined traffic solver, reachable control witness, clearance or gradient is certified. The readable report is `sampled_insertion_findings.md`; source locations are in `sampled_insertion_source_audit.md`.

1. Completed transfer and newborn metadata are separate

At regular update t_{k+1}, let U^{k+1} be the completed own population/count state, including all accepted transfers during $[t_k, t_{k+1})$. Pre-existing restrictions are part of that interval's traffic problem. For directed generating stream e, let $\Delta m_e^k \geq 0$ be actual continuous transfer mass, H_e^k its cumulative mass and $m_* > 0$ the fixed threshold unit.

$$H_e^{k+1} = H_e^k + \Delta m_e^k, \qquad b_e^k = \left\lfloor H_e^{k+1}/m_* \right\rfloor - \left\lfloor H_e^k/m_* \right\rfloor. \tag{1}$$

With remainder $z_e^k \in [0, 1)$, the exact bookkeeping is

$$b_e^k = \left\lfloor z_e^k + \Delta m_e^k/m_* \right\rfloor, \qquad z_e^{k+1} = z_e^k + \Delta m_e^k/m_* - b_e^k. \tag{2}$$

The floor counts descriptors, never rounded physical traffic. For each of the b_e^k newborns, the selected reset reads

$$t_b = t_{k+1}, \quad \ell_b = \ell_{\text{target}}, \quad x_b = x_{\text{entry,e}}, \quad v_b^{\text{des}} = v_{\text{donor,e}}^{k,\text{request}}, \quad h_b = N_{\ell_b}^{k+1}(x_b). \tag{3}$$

Here N is the own count field with the declared gauge and accepted source accounting. Ramp $x_{\text{entry,e}}$ is the physical merge; discretionary entry uses the declared target-section coordinate. These are E1 adaptations. The first realized speed still obeys source CM:

$$v_b = \min\{v_b^{\text{des}}, v_b^{\text{dwn,own}}\}. \tag{4}$$

This relation does not independently supply the joined downstream-speed estimator.

Let $\mathcal{D}^+$ be the retained old/new descriptor list after reset and $\mathcal{R}$ the metadata reset. It satisfies

$$\mathcal{R} : (U^{k+1}, H^{k+1}, \mathcal{D}^-) \longmapsto (U^{k+1}, H^{k+1}, \mathcal{D}^+), \qquad M(U^{k+1})_{\text{after}} = M(U^{k+1})_{\text{before}}. \tag{5}$$

M includes outside waiting, the approach and all road lanes. Paired internal transfers cancel; only offered arrivals and sink exits change total population. Completed exit histories are unchanged. Equal own inputs/histories through t_{k+1} yield equal newborns through that time; no future reference is used. This is conditional reset causality, not existence of the preceding solve.

2. Count attachment must be a value of the traffic field

For same-lane newborns i, j at the same update and location,

$$x_i = x_j \quad \Longrightarrow \quad h_i = h_j = N_\ell^{k+1}(x_i). \tag{6}$$

Different speeds or physical parameters do not imply different N values at one point. For neighboring distinct positions and jam density κ, admissibility requires

$$0 \le N_\ell^{k+1}(x_i) - N_\ell^{k+1}(x_j) \le \kappa(x_j - x_i), \qquad x_i < x_j. \tag{7}$$

In particular, artificial label spacing at zero spatial separation violates the upper bound. Descriptor multiplicity is not extra population. Identical complete signatures may be grouped losslessly; differing signatures are kept. Equal initial labels do not establish equal future paths.

Existing paths retain the preceding source correction. On a regular sourced segment, with boundary x_0, net lateral source s and $\dot{\phi} = v$,

$$\frac{d}{dt} N(t, \phi(t)) = q(t, \phi) - v\rho(t, \phi) - \int_{x_0}^{\phi(t)} s(t, y)\, dy. \tag{8}$$

Zero relative passing removes the first two terms, not the source integral. This does not choose attachment for an atomic source on the path.

3. One local active trace explains the service mechanism

Consider only a source-free segment with triangular fundamental diagram, positive free/wave speeds u, w, jam density κ, and an active zero-passing restriction moving at prescribed $0 \le v < u$. An upstream saturated trace and a downstream free trace satisfy

$$Q(\rho) = \min\{u\rho, w(\kappa - \rho)\}, \qquad Q(\rho_U) - v\rho_U = Q(\rho_D) - v\rho_D = 0. \tag{9}$$

Selecting those branches gives

$$\rho_U = \frac{w\kappa}{w + v}, \quad q_U = \frac{vw\kappa}{w + v}, \quad \rho_D = q_D = 0, \qquad \frac{dq_U}{dv} = \frac{w^2\kappa}{(w + v)^2} > 0. \tag{10}$$

The zero downstream trace is the void adjacent to the moving restriction. This is *not* a global fixed-section service cap or new node rule. All node receiving and later exits must follow the same wave solution. For diagnostic values $u = 30$, $w = 5$ m/s, $\kappa = 0.15$ veh/m and $v = 10$ m/s, $q_U = 0.5$ veh/s versus road capacity $uw\kappa/(u + w) = 9/14$ veh/s. These are hand-selected local inputs, not calibrated or reachable metering cases.

4. Simultaneous prescribed paths have a wider analytical route

For a homogeneous source-free road with prescribed paths $\phi_j(t)$, the relative road capacity at observer speed ξ is

$$R(\xi) = \sup_{0 \le \rho \le \kappa} \{Q(\rho) - \xi\rho\}, \qquad -w \le \xi \le u. \tag{11}$$

Along a bottleneck, its declared maximum relative passing rate $r_j(t)$ reduces that cost. Applicable coincident restrictions combine by a minimum, not by an additive population or capacity debit. For compatible data on boundary Γ, the variational reference reads

$$N(t,x) = \inf_{\gamma : B \in \Gamma \to (t,x)} \left\{ N(B) + \int_{t_B}^{t} R_{\mathcal{D}}(\dot{\gamma}(\tau), \tau, \gamma(\tau))\, d\tau \right\}. \tag{12}$$

$R_{\mathcal{D}}$ is the road cost reduced on the applicable prescribed curves. All distinct curves remain. V03, Theorems 1–3, supplies scoped validity and boundary consistency conditions; V02 discusses graph representations and their errors. A finite graph is not declared exact or adopted here.

The source pairwise-CRP step bound includes

$$\Delta t \le \frac{x_{i+2} - x_i}{w + v_i}. \tag{13}$$

A triple coincidence makes this bound zero when $w + v_i > 0$. That demonstrates the limit of the pairwise construction, not nonexistence of the multi-path variational solution.

5. The remaining feedback is an actual qualification obligation

In the full task the paths are not prescribed independently of traffic: CM needs the own downstream speed; the paths constrain that traffic; node receiving and accepted sources must come from the same solution. V03 Section 5 distinguishes this endogenous case. Neither the displayed infimum nor a name for its solver proves that the joined update is determinate.

The sampled reset is conservative and causal. Prescribed-path continuation has positive source support within its assumptions. The full own-speed/node/source update and eventual clearance remain partial; complete TST leverage, reduction necessity and useful gradients are not established. Exact information remains permitted; no recovery approximation, position perturbation, changed clock, replacement solver or experiment is adopted.

Sources. A01: supplied journal Sections 2.2 and Appendix A. S02: RR-2003-2 Sections 2–4. V01: RR-2003-3 Sections 1–3. V02: RR-2003-7 Sections 2 and 4.1. V03: Daganzo (2006), NHM 1(4), Sections 2.1 and 5.